

HELSINKI KAISANIEMI, Finland

WMO: 029780

Lat: 60.1753N

Lon: 24.9447E

Elev: 4

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-15.2	-21.0	0.6	-18.0	-17.6	0.8	-14.5	9.2	-1.2	8.4	-1.3	3.2	40	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	25.9	19.9	24.1	18.9	22.6	17.6	21.0	24.5	19.9	23.2	18.5	21.8	3.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.4	23.3	18.5	13.4	22.3	17.0	12.2	20.6	60.7	24.5	56.9	23.1	52.4	22.0	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.7	-20.0	27.8	4.4	2.5	-23.1	29.6	-25.7	31.1	-28.2	32.5	-31.4	34.3		
			-20.2	21.3	4.3	1.6	-23.3	22.5	-25.9	23.5	-28.3	24.4	-31.4	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-4.6	-3.2	-0.3	4.8	11.1	14.9	18.4	17.0	12.7	6.6	3.2	-0.2
	DBStd	8.57	5.59	5.45	3.63	3.02	3.81	2.94	2.93	2.57	2.96	3.36	3.89	4.77
	HDD10.0	1974	452	368	321	158	31	2	0	0	11	111	203	317
	HDD18.3	4282	710	602	579	406	226	108	36	56	169	363	453	576
	CDD10.0	792	0	0	0	3	64	150	259	217	93	6	0	0
	CDD18.3	59	0	0	0	0	1	6	37	14	0	0	0	0
	CDH23.3	236	0	0	0	0	4	19	170	43	0	0	0	0
	CDH26.7	26	0	0	0	0	0	1	22	3	0	0	0	0
	WSAvg	3.6	3.8	3.7	3.8	3.3	3.4	3.5	3.3	3.3	3.5	3.8	4.0	4.3
	PrecAvg	603	44	32	30	34	36	52	56	73	60	68	66	50
Precipitation	PrecMax	838	84	100	57	71	73	125	152	165	155	177	170	128
	PrecMin	360	0	0	4	7	5	11	2	1	17	16	4	6
	PrecStd	102	22	23	15	19	20	27	40	40	33	36	31	27
	PrecStd	102	22	23	15	19	20	27	40	40	33	36	31	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	6.2	9.0	16.6	23.6	25.6	28.7	27.0	20.4	14.2	10.6	8.5
		MCWB	4.4	3.1	5.6	9.4	15.2	18.1	20.2	20.9	16.3	11.5	9.4	7.5
	2%	DB	4.2	4.1	6.9	14.1	21.5	23.2	26.9	24.3	18.8	12.9	9.3	6.9
		MCWB	3.5	2.7	3.5	8.7	14.2	16.7	20.1	19.8	15.7	11.2	8.5	5.8
	5%	DB	3.4	3.1	5.2	11.7	19.6	21.7	25.2	22.7	17.8	12.2	8.5	5.9
		MCWB	2.7	2.2	2.9	7.2	13.5	15.8	19.8	18.1	15.2	10.8	7.6	4.9
	10%	DB	2.2	2.3	3.8	9.8	17.7	20.2	23.7	21.2	16.9	11.3	7.8	5.0
		MCWB	1.7	1.7	2.1	5.8	12.3	14.9	18.8	17.1	14.6	10.0	7.0	4.2
	0.4%	WB	4.7	3.7	5.6	10.2	17.3	19.8	22.2	21.9	17.2	12.6	9.8	7.8
		MCDB	4.9	4.9	8.3	15.1	21.1	23.1	26.1	24.9	18.9	13.6	10.4	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.6	3.0	4.1	8.8	15.5	18.0	21.5	20.4	16.5	12.0	8.7	6.0
		MCDB	4.1	4.0	5.9	13.1	19.2	22.3	25.1	23.7	18.2	12.7	9.3	6.7
	5%	WB	2.8	2.3	3.3	7.7	14.2	16.6	20.7	19.2	15.8	11.1	7.8	5.1
		MCDB	3.4	2.9	4.8	11.3	18.5	20.5	24.2	21.9	17.3	11.9	8.3	5.7
	10%	WB	1.8	1.7	2.4	6.5	12.8	15.5	19.5	17.8	15.0	10.2	7.1	4.3
		MCDB	2.2	2.3	3.7	9.2	17.5	19.0	23.0	20.5	16.5	11.1	7.7	5.0
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.1	5.1	6.3	7.2	6.8	6.7	6.1	5.4	4.7	3.4	3.7
		MCDBR	3.2	3.6	6.2	9.3	9.6	8.8	8.1	7.7	6.2	5.1	3.3	3.6
		MCWBR	3.1	3.0	4.0	5.2	4.9	4.5	3.8	3.8	3.9	4.0	3.3	3.6
	5% WB	MCDDBR	3.2	3.4	5.4	8.4	8.7	8.1	7.2	6.4	5.6	4.1	3.3	3.4
		MCWBR	3.1	3.0	3.8	5.1	4.9	4.6	3.8	3.7	3.9	3.9	3.4	3.4
Clear-Sky Solar Irradiance	taub		0.280	0.313	0.335	0.360	0.358	0.356	0.378	0.372	0.345	0.334	0.307	0.265
	taud		2.141	2.192	2.247	2.340	2.414	2.477	2.438	2.454	2.519	2.394	2.216	2.091
	Ebn at Noon		473	669	785	838	869	876	844	820	786	653	436	333
	Edh at Noon		43	72	94	103	103	99	100	92	73	60	41	30
	RadAvg		0.25	0.82	2.31	3.85	5.25	5.46	5.27	4.08	2.49	1.08	0.30	0.15
All-Sky Solar Radiation	RadStd		0.04	0.16	0.30	0.40	0.57	0.43	0.48	0.42	0.32	0.17	0.05	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-140	N/A	N/A	N/A

Nomenclature: See separate page